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Building the algorithm commons: Who discovered the algorithms that underpin computing in the modern enterprise?

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Abstract

Research Summary: The reach of the modern enterprise relies on the power of information technology (IT) tools such as sensors, databases, and machine learning. But tool improvements must be fueled by increased computing power (e.g., faster hardware) or getting more productivity from existing systems (e.g., through better computer algorithms). New research has uncovered that this second source, algorithm progress, is more important than previously realized—sometimes orders of magnitude more important than hardware—and thus could be an important technological stepping-stone to give competitive advantage to a country's firms. Analyzing this "Algorithm Commons" reveals that the United States has been the largest contributor to algorithm progress, with universities and large private labs (e.g., IBM) leading the way, but that U.S. leadership has faded in recent decades.

Managerial Summary: Companies are increasingly tackling problems with big data and sophisticated analysis techniques (e.g., Machine Learning). To meet the increased computational demands of these approaches, the capability of computers must improve. One important

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technique for doing this is to redesign algorithms, the recipes that computers follow to perform calculations. As a result, firms that develop better algorithms, or get access to them first, can get important advantages over their competitors. This paper shows that it is U.S. corporations and U.S. universities that have produced most of the important algorithm improvements, and thus suggests that better algorithms may have been a source of advantage for U.S. multinationals.

KEYWORDS

algorithms, computing, digital economy, information technology, public goods

1 | INTRODUCTION

Modern enterprises rely strongly on information technology to organize their data and analytics (Brynjolfsson & McElheran, 2019), improve productivity (Brynjolfsson & Hitt, 1996; Thompson, 2017), and manage large multinational operations (Bloom, Sadun, & Van Reenen, 2012; Mauri & Neiva de Figueiredo, 2017; McDermott, Mudambi, & Parente, 2013; Stallkamp & Schotter, 2019). But staying at the forefront of IT requires exponential increases in computing power to feed new functionality, big data, and analytical tools (Thompson, Greenewald, Lee, & Manso, 2020; Thompson, Ge, & Manso, 2020; Thompson & Spanuth, forthcoming). Some of this increase comes from improved computer chips (although less as Moore's Law slows (Thompson & Spanuth, forthcoming)), but new evidence suggests that algorithmic improvement has made a larger contribution than was previously realized (Sherry & Thompson, 2020).

Algorithms are one of the most important parts of computing, alongside hardware and software (Leiserson et al., 2020). Algorithms organize computations, converting a goal (e.g., sorting a list) into a series of steps that the computer will perform to reach that goal. Many algorithms are so embedded into computer systems that users and programmers never notice them. For example, a coder in a programming language like Python or R might call a standard function (e.g., "sort" or "matrix multiplication") without realizing that, behind the scenes, the designer of that function built it using an algorithm that ensured that the calculation would be done efficiently.[1] Despite going unseen, algorithmic improvements can have a transformative impact on calculations, for example by taking a computation that was so computationally expensive that it was impossible on even the largest supercomputers and making it tractable. This is, for example, what quantum computing algorithms are expected to do with code-breaking (Ekert, 1997; Roetteler & Svore, 2018).

Algorithm improvement is analogous to the economic concept of productivity improvement. Just as a productivity improvement allows a firm to produce more output for a given set of inputs, an algorithmic improvement allows it to tackle a bigger, harder problems for the same computational budget. Recent work by Sherry and Thompson (2020) has shown that improvements in algorithms have been large: rivaling or exceeding the decades-long improvements in computer hardware coming from Moore's Law for many problem types.

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It has long been argued that "the main driver of the digital economy is the continued exponential improvements in the cost-performance of [Information and Communications Technology], mainly microelectronics, following Moore's Law" (Sturgeon, 2019). The new findings about algorithms suggest that they may have subtly been one of the key drivers of the digital economy, and thus that firms with better access to cutting-edge algorithms may have had a significant competitive advantage.

In the real world, there are many examples where algorithms provided so much advantage that they propelled large multi-national enterprises.[2] Famously, Google relied on its PageRank algorithm to overcome other search engine competitors. Another example is Uber, which operates its taxi service in more than 65 countries. Routing algorithms are central to Uber's business, since the company must constantly estimate how long trips will take and how much to charge customers as traffic patterns evolve. Significant improvements in their graph processing algorithms now allow Uber to update time-of-arrival calculations across millions of nodes in just a few minutes, whereas previously such updates took 12+ hours (Nguyen, 2015).

In other cases, the effect of algorithms is less visible: suffusing software with additional performance that is only observed by customers through better firm performance. Akamai Technologies is an MIT-spinoff that has become the world's largest platform for distributing content over the internet. Their patented routing algorithms direct user requests for content to the optimal servers, ensuring that delivery is fast and along the most-efficient routes. Akamai handles 30% of all web traffic, serving clients like Google, Apple, and Netflix and processing more than two trillion Internet transactions every day (Akamai, 2020).

Another subtle example is Netflix. While video content is clearly their main business, delivering that content through overcrowded or low-bandwidth internet lines can be challenging, for example in countries with poorly-developed broadband. Netflix has a dedicated team of developers working on these issues, and has found algorithm improvements that allow it to deliver content of the same quality using half the bandwidth (Manohara, Moorthy, Cock, Katsavounidis, & Aaron, 2018).

One complication to the usage of algorithm improvement for firm advantage is that algorithms are typically public goods because they are non-rivalrous information. Thus, once an algorithm is known, it can be used by all. This public good aspect of algorithms is partly enforced by law, since the patenting of algorithms is limited (Rapacke, 2020). Algorithm improvements also come predominantly from universities, presumably because they are the source of so much basic R&D (Arora, Belenzon, & Patacconi, 2018; National Science Board, 2016), further reinforcing algorithms' place in the Digital Commons because of the norms of science (Merton, 1973). Thus, strategically, relying on algorithms for firm advantage is more akin to relying on open source software.

As previous work has shown, relying on the digital commons can provide significant (if often uncounted) benefits for firms (Brynjolfsson & Saunders, 2009; Greenstein & Nagle, 2014; Sturgeon, 2019). And, since the diffusion of knowledge is often geographically constrained (Allen, 1984), this also means that firms from particular countries may be advantaged when they can build upon a stronger algorithmic foundation. This paper presents the first piece of a larger effort to quantify how much better algorithms enhance firm performance. In it, we analyze who produced the Algorithm Commons and thus who stands to benefit from it.

We find that the short answer to "who is building the Algorithm Commons" is: the United States. 38% of all algorithmic improvements have come from researchers born in the United States and 64% of improvements were made by researchers working in United States when they made the discovery. Outside the U.S., we find that being a richer country, either in terms of

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GDP per capita or total GDP, is predictive of making more algorithm contributions. The size of a country's population is not meaningfully related to producing more algorithm contributions, except insomuch as this manifests through more GDP. This suggests that algorithm development may exhibit the "lost Einsteins" problem seen in the creation of patentable inventions (Bell, Chetty, Jaravel, Petkova, & Van Reenen, 2019), wherein those with natural talent never get to benefit the world because of a lack of opportunity.

We also examine the institutions that researchers were at when they made their discoveries. 82% of discoveries were made at public/not-for-profit institutions such as universities or in government. Unsurprisingly, we find that the top public institutions are those universities that rank highest in computer science: Stanford, MIT, UC Berkeley, and Carnegie Mellon. Collectively, these four institutions account for one-sixth (17%) of all algorithm improvements. This finding highlights an important contribution by universities to the public that is usually missed.

Overall, the private sector contributes less to algorithms than the public/not-for-profit sector. Nevertheless, several private sector labs are large contributors, particularly: IBM, Bell Labs, and the Rand Corporation. Indeed, IBM is the single largest contributor to algorithmic improvement, having made 6.3% of all the discoveries. This raises the interesting question of why firms are investing in algorithms, since public goods provide much weaker incentives (Arrow, 1962; Samuelson, 1954). As we argue in the theory section, it seems likely that these discoveries arise because firms want the technical improvement as a user (Von Hippel, 2007; Von Hippel, 2009) or because such expertise is part of a larger strategic choice to build absorptive capacity (Cohen & Levinthal, 1990).

Finally, we observe that contributions to the Algorithm Commons have mimicked larger political-economy trends, with contributions coming predominantly from the United States in the early post-war period, but with that dominance fading in recent decades as Europe and Asia have grown in importance.

2 | THEORY AND HYPOTHESES

There has been longstanding interest in how firms benefit from I.T. and how it allows them to build large, multinational firms. For example, Stallkamp and Schotter (2019) discuss how improved connectivity and digitalization have allowed for a flowering of multinational platform firms. McDermott et al. (2013) argue for the importance of modularity in multinationals' architecture and geographic dispersion, and note that I.T. is an area with substantial modularity.

The question of how firms benefit specifically from algorithms is more complicated, however, since they are public goods. As the early work (Arrow, 1962; Samuelson, 1954) emphasized, the production of public goods results in positive spillovers to others, which leads to insufficient private incentives for production and thus we should expect underprovision.

At the same time that public goods are likely to be underprovided, when they do happen they are likely to provide substantial gains for firms in their vicinity. Empirical work testing for positive spillovers in innovation and discovery, the types of public goods that we focus on, found that "R&D spillovers are both prevalent and important" (Baldwin & Von Hippel, 2011; Griliches, 1992). These positive spillovers have been shown to have important effects on firms, increasing productivity (Czarnitzki & Kraft, 2012; Nagle, 2014), providing strategic advantages (Harhoff, Henkel, & Von Hippel, 2003), and building innovation capabilities (Altman, Nagle, & Tushman, 2014; Golovko & Valentini, 2014; Govindarajan & Ramamurti, 2011; Trachuk & Linder, 2019). Thus, there is good reason to believe that algorithm development could have

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positive spillovers, providing a country-specific advantages for firms of that country (Mauri and de Figueiredo, 2017).

In recent decades, there has been a flowering of other types of public goods in information technology, which can be informative about how algorithms might affect firms and what motivates their provision. These public goods in I.T. can take many forms, for example information provision on Wikipedia and open source software projects such as Linux or the Berkeley Software Distribution. These "voluntary 'crowdsourcing' efforts of engineers and technologists interested in creating free resources for their communities" (Sturgeon, 2019) have engendered considerable attention to what motivates individuals, firms, and countries to contribute to public goods. At the individual level, many theories have been proposed. One, is that contributors get intrinsic benefits, particularly if their contribution meets the needs that they have at work (Franke & Von Hippel, 2003; Shah, 2006), or if it contributes to a larger mission that they believe in (Franke & Von Hippel, 2003; Gelernter, 1998; Kuan, 2001; Lakhani & Von Hippel, 2003; Lerner & Tirole, 2002; Raymond, 1999; Weizenbaum, 1976). Extrinsic benefits are also believed to play a role in motivating contributions, for example because of the status and reputation that individuals achieve (Belenzon & Schankerman, 2015; Lindenberg, 2013; Raymond, 1999), the social recognition and awards in public projects (Gallus, 2017) and through benefits to career development (Belenzon & Schankerman, 2015; Hann, Roberts, Slaughter, & Fielding, 2002; Lerner & Tirole, 2002). These latter effects may be particularly important for contributions coming from academia, where the norms of status and recognition favor the spread of knowledge (Merton, 1973).

Individual contributions may also be more attractive to make when the design of a project is modular, which makes changes easier to make (Baldwin & Clark, 2004; Parnas, 1972)—which is perhaps why these are common characteristics in public good projects (Benkler, 2002) and why the Algorithm Commons may be particularly attractive. Thus, at the individual level, theory suggests that contributions will be primarily by those either using the algorithms themselves (user-innovators) or by those who will gain status, career improvement or other personal benefits from discoveries (e.g., academics).

Even if there are clear reasons for individuals to contribute to algorithm development, why do firms do so? Open-source software provides guidance on this question. More than half of firms use or contribute to open-source software (Black Duck Software, 2014) and more than half of computing devices use it (e.g., Android) (Yarow, 2013). Certainly, it would be tempting for firms to free-ride on the open-source software produced by others. For example, Nagle (2019) found that a change of procurement policy by the French government led to thousands of new open-source software contributions per month, which would have cost software companies $20 million per year to replicate. And yet, despite this, billions of dollars are poured into the open-source software ecosystem by venture capitalists (Black Duck Software, 2014; Forest, 2014; Hamilton, 2014; Lundun, 2014) and by firms. For example, IBM has invested close to $4 billion in open-source software (Moore & Ferris, 2020).

The most compelling answer to why firms contribute seems to be that they themselves benefit. For example, Nagle (2018) looked at contributions to Linux, and found that firms who contribute became 11% more productive than matched firms that did not. Contributions to open-source software may also be part of a larger R&D strategy designed to maintain technology leadership (AI MarketPlace, 2018; Capek, Frank, Gerdt, & Shieds, 2005; CBInsights, 2020), may help firms build better products by allowing them to collaborate with others (Von Hippel & Krogh, 2003) (Daffara, 2007; Moore & Ferris, 2020), or may create substantial value through the sales of complementary goods, such as consulting (Daffara, 2007; Munga, Fogwill, & Williams, 2009). These same trends seem to also play out in open source hardware (Gambardella & Von Hippel, 2018).

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It may also be that, while eventually algorithmic improvements become public and broadly used, they provide an initial benefit to firms involved in their development. This adoption head-start could arise from better awareness that the improvement happened, or it could reflect "absorptive capacity" (Cohen & Levinthal, 1990; Von Hippel & Krogh, 2003) wherein firms that develop an algorithm (or a related one) have expertise that allows them to better implement it. If other parts of the world have less algorithmic absorptive capacity than the U.S., as one might surmise from their overall contributions to algorithms, then this home-country complementary asset could be an advantage to U.S. firms as they expand internationally (Mauri & Neiva de Figueiredo, 2017; Stallkamp & Schotter, 2019).

The arguments for firm provision of public goods seem compelling in some cases, perhaps especially so for open-source software. Nevertheless, the foundational arguments of Samuelson (1954) and Arrow (1962) remain paramount, and thus we would expect algorithmic provision to happen predominantly in the public/not-for-profit sector, as has been happening increasingly with basic R&D of all types (Arora et al., 2018; National Science Board, 2016).

Finally, we consider why it is that some countries contribute more to the Algorithm Commons than others. One suggestion, from the international relations literature, is that these contributions would be part of a larger deal between global hegemons and others: countries in positions of power get to set international rules, but they must also provide public goods like security that ensure the buy-in of others (Kindleberger, 1986; Webb & Krasner, 1989). In this case, we would expect the leading countries in each era to contribute more to the Algorithm Commons.

Another possibility, motivated by economics, is that algorithm development is a good innovation choice from a sub-optimal choice set. To understand this, consider that more-developed countries must innovate afresh to improve productivity, while less-developed countries can catch-up more easily by adopting known technologies (Sala-i-Martin, 2006). Faced with having to innovate to improve, more-developed countries must invest. Even if the private incentives to invest in public goods are less than their full benefits, it may still make sense to invest in algorithm development if the returns are comparable to other options. When such development is funded by the state, for example the U.S. government, it may also be that many of the externalities are captured, for example because some global benefits from improvements are captured by U.S. companies present in other geographies. Thus, an economic view of optimal investments makes the same prediction as the international relations view: leading countries of each era should contribute more to the development of the Algorithm Commons.

3 | METHODOLOGY

3.1 | Sample and data collection

A full description of the algorithm data and how it was gathered can be found in Sherry and Thompson (2020), but the following summarizes it: 57 textbooks and 1,141 research papers were used to assemble the first comprehensive look at algorithms and how they have developed. The data was divided into algorithm families: sets of algorithms that all solve the same problem. A family was included in the data if at least one textbook author felt it was important enough to include and if the problem statement, as well as the algorithm solutions, were provably exact.[3] In total, there are 128 algorithm families in the data. Within each family, only those algorithms that either were the first in that family or that substantially improved on the state-of-the-art were[4] considered. There are 310 algorithm papers that make this large a contribution.

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For each algorithm in the Sherry and Thompson (2020) data set, we search for the original paper that described the discovery. In some cases, this is not available, either because it originates in antiquity (e.g., algorithms developed by the Greeks) or because an initial implementation is the naïve/brute-force approach[5] and thus is never announced. We are able to source the original paper for 60% of algorithms. Using these, we extract information about the algorithm discoverers and their organizations. In particular, we gather: the author(s), their institution(s) and their address(es). Importantly, we are interested in the location of the author(s) when the work was done. When that information is not explicitly listed on the paper, we search other resources to find it, including Microsoft Academic Search, Scopus, Web of Science, and Wikipedia (in that order). We are able to source institution information where the author did their work in for 88% of those algorithms where there are original papers. We disambiguate the resulting locations and institutions of the authors manually and then classify them, for example into private vs. public/not-for-profit.

We also gather the birth country for each author in our data, although this process is more involved. Of the 405 authors, we were able to find explicit mentions of the birth location of 41% of them in public records, such as on their personal website or on Wikipedia. For the remaining 59%, we use other methods in the following order: based on the author's CV, we assume that they were born in the country of their earliest educational institution (high school or undergraduate degree) (37%), we email the discoverers directly (3%), and we use surname origin databases (Ancestry Corporate, 2020; House of Names, n.d.), Wikipedia, (Forebears, 2020), and (Name Origin Research, 2017) to find the most-likely birth country (19%).[6]

For the countries that the authors were either born in, or worked in, we gather the most recent year country-level information on GDP from the World Bank (The World Bank, 2019a, 2019b, 2019c) and on population from Worldometer (Worldometer, 2020).

3.2 | Outcome

The key outcome that we consider is the number of key algorithm papers that are produced. In order to apportion a paper across multiple authors or multiple institutions, we divide credit for a paper as follows:

$$Contribution_{a,i} = \frac{1}{\|A\|} \cdot \frac{1}{\|I_a\|}$$

That is, the contribution made by author a, in group of authors A, who works at institution(s) $i \in I_a$ is an equal division. For example, if a paper had two authors, Lynne at TechU and Jonathan with a cross-appointment at CollegeU and StateU, then the relative contributions would be: $\left(Lynne_{TechU}, Jonathan_{CollegeU}, Jonathan_{StateU}\right) = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right)$. In this way, we avoid double-counting contributions, while still giving credit to the various institutions and countries of the authors.

3.3 | Analysis plan

Since this is the first presentation of this data on this foundational topic underpinning modern computing, we focus primarily on providing summary statistics that describe the building of the

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Algorithm Commons. Nevertheless, at the country level, we do estimations of the relative algorithm contributions made by country c based on its GDP and population characteristics:

$$Contribution_c = \beta_0 + \beta_1 \cdot GDPperCapita_c + \beta_2 \cdot Population_c + \beta_3 \cdot GDPperCapita_c \cdot Population_c + \beta_4 \cdot USA + \varepsilon$$

We add an explicit dummy variable for contributions by the USA, reflecting its outsized role in algorithm development.

To avoid considering only intensive variation amongst those doing algorithmic developments, we augment our data with countries that did no algorithmic developments until we have the 70 largest countries in the world by GDP.

Notice that our analysis is cross-sectional, rather than a panel. This raises the possibility that we may have omitted variable bias in our regression, for example due to differential GDP growth rates, which might be expected to be correlated with algorithm contribution and GDP level. However, as Table 1 will show, our models that include the USA dummy explain nearly all variation, for example, $R^2 = 0.987$ for regression (4). This is important since the size of potential omitted variable bias is a function of unexplained variation (Hanck, Arnold, Gerber, & Schmelzer, 2019) as expressed in the residual standard error, σ_ε:

$$\left(\hat{\beta}_i - \beta_i\right) \xrightarrow{p} \rho_{X\varepsilon}\left(\frac{\sigma_\varepsilon}{\sigma_X}\right)$$

But, since R^2 is nearly one, the size of our residual standard error, σ_ε, is small compared to σ_X. For example, for *GDP per capita* in regression (4), it is 0.6 $\left(=\frac{1.4}{2.4}\right)$. Thus, even if another variable was correlated as highly with the errors as GDP per capita is with algorithm contributions ($\rho = 0.23$) then the bias would only be relatively small (0.14). For more realistic correlations, one would not expect bias to be a concern. This is a special case of the more-general phenomenon of better explanatory power reducing the potential for confounding (VanderWeele & Ding, 2017). Importantly, this result only speaks to the combined direct and indirect effects of these variables (Baron & Kenny, 1986). That is, it would be consistent with our empirical findings that, for example, having more computer science or mathematics faculty is directly related to the number of algorithmic contributions produced by that country, but that variation in those such variables is largely a consequence of higher GDP per capita or having a larger population, and thus is an indirect effect of it.[7]

4 | RESULTS & DISCUSSION

In Figure 1, we highlight the starkest findings from our analysis. In particular, we see that the United States has been, overwhelmingly, the largest contributor to algorithm development with 38% of all discoverers born there and 64% of all discoverers working there. This reinforces a long-standing argument that the U.S. has been able to attract many of the best and brightest from around the world to work at U.S. institutions (Hoffman, Shoss, & Wegman, 2020). 42% $\left(=\frac{64-38}{100-38}\right)$ of discoverers from other countries who made major algorithm contributions did so while they were working in the United States.

The concentration of algorithm research work in the United States, including by foreign-born scientists, echoes a trend seen in R&D spending more generally. A significant share of

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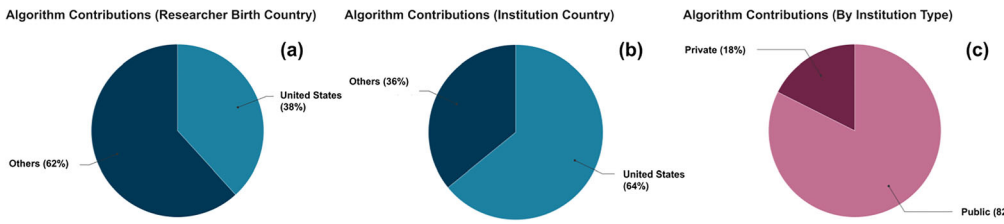

FIGURE 1 Algorithm contribution summary

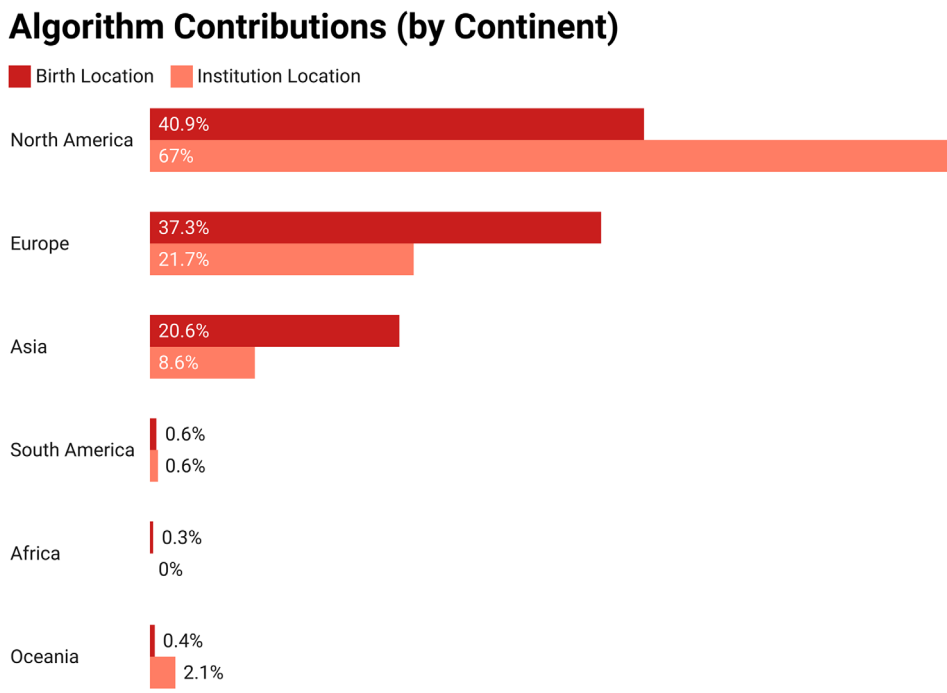

FIGURE 2 Algorithm contributions (by continent)

large international companies perform their R&D in the United States. Swiss firms, for example, performed almost as much private-sector R&D in the U.S. (Boroush, 2020) as all R&D performed in Switzerland (OECD iLibrary, 2020). For other countries, the corresponding ratio is lower but still substantial, for example, 23% for the United Kingdom and Netherlands, and 16.3% for France (OECD iLibrary, 2020).

It is thought that this attraction of foreign firms to do R&D in the U.S. comes from its role as a attractor of global technological talent and cutting-edge technology (Dunning & Narula, 1995). It is thus likely that U.S. algorithm development is both a cause and effect of this broader R&D concentration in the U.S.

We also see that 82% of algorithm contributions were made in the public/not-for-profit sector, as we would expect since they are public goods.

Figure 2 shows the large disparity in algorithm contributions between the developed world and developing world. Despite the distribution of the world's population, North America (5%), Europe (10%), Asia (60%), South America (6%), Africa (17%), and Oceania (<1%)[8] (United

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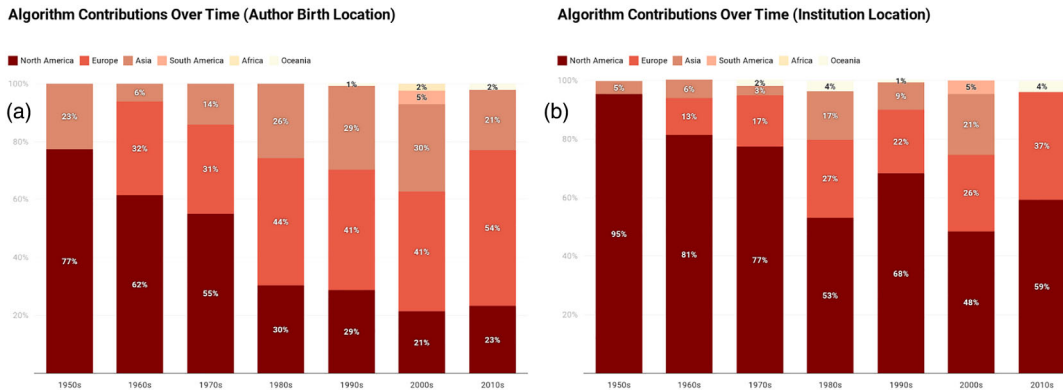

FIGURE 3 Algorithm contribution over time

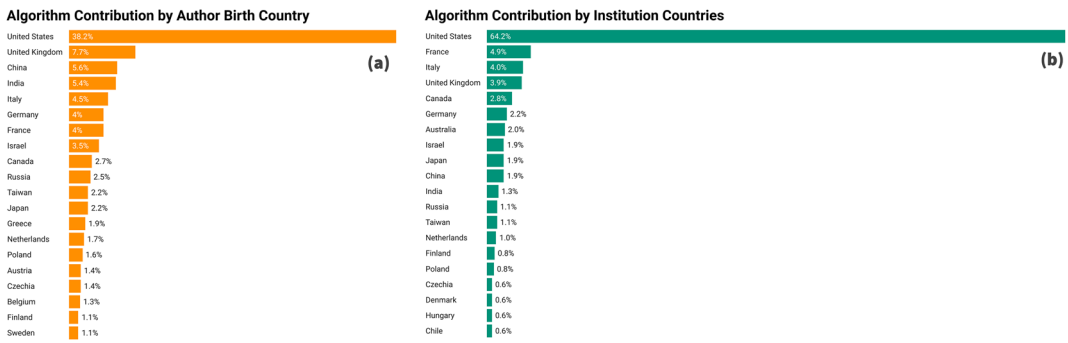

FIGURE 4 Algorithm contribution by country

Nations World Population Prospects, 2020), algorithm contributions come overwhelmingly from institutions in North America and Europe (89%) and from those who were born there (78%). This shows that it is not simply that these locations are gathering talent from around the world, but that such talent is only rarely being developed outside these places, despite the large share of world population that live there. This provides additional evidence supporting the view from (Bell et al., 2019) of "lost Einsteins," whose natural talent never benefits the world because they lack the opportunity to develop it.

Figure 3 continues the analysis of contributions by continent, examining how these are changing over time.[9] Figure 3a reveals that the disparities in birth country have significantly diminished in recent decades, although they remain far from representative. Figure 3b, however, shows a far smaller diminishment in the dominance of North American and European institutions in these contributions. It shows that in recent years talent is being more evenly sourced from around the world, but that the locations of work remain concentrated in North America and Europe.

Figure 3a also validates theory predictions about which countries will contribute most to public goods, showing American dominance in the post-WW2 period, and the gradual rise of Asia in recent decades.

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TABLE 1 Determinants of algorithm contributions at the country level

	Dependent variables: Number of algorithm improvements			
	Birth country (1)	Birth country (2)	Institution country (3)	Institution country (4)
Constant	−0.413	0.312	−1.020	0.197
	(1.546)	(0.441)	(2.506)	(0.296)
GDP per capita ($10,000)	1.005**	0.297**	1.371*	0.184**
	(0.447)	(0.130)	(0.724)	(0.087)
Population (100,000,000)	0.612	0.368*	0.259	−0.151
	(0.684)	(0.195)	(1.109)	(0.131)
GDP per capita * Population	0.093	0.467*	−0.078	0.549***
	(0.850)	(0.243)	(1.378)	(0.163)
United States		66.992***		112.387***
		(2.449)		(1.644)
Observations	70	70	70	70
R^2	0.093	0.928	0.054	0.987
Adjusted R^2	0.052	0.923	0.011	0.986
Residual *SE*	8.359 ($df = 66$)	2.381 ($df = 65$)	13.544 ($df = 66$)	1.598 ($df = 65$)
F statistic	2.257* ($df = 3; 66$)	207.893*** ($df = 4; 65$)	1.254 ($df = 3; 66$)	1,235.869*** ($df = 4; 65$)

*Significant at the 10 percent level
**Significant at the 5 percent level.
***Significant at the 1 percent level.

Figure 4 provides additional details of country contributions, showing the top 20 countries for (a) author birth country and (b) author institution country. Table 1 considers the relationship between these contributions and each countries' GDP and population.

As Figure 4 and Table 1 make clear, the United States is unique in its contributions to the Algorithm Commons. Taking that into account with a dummy variable and then focusing on regressions (2) and (4) reveals that GDP per capita and total GDP (GDP per capita × Population) both statistically significantly predict algorithm contributions, and collectively represent a moderate-size effect. In contrast, while population is predictive of birth country contributions, a very large increase (100 million inhabitants) is needed to approximately equal the effect of an additional $10,000 in GDP per capita, so the size of the effect is not economically meaningful, even if the coefficient is statistically significant. In terms of the location of work, regression (4) shows that the point estimate for population is negative, indicating that having a more populous country has, if anything, a negative effect on producing algorithmic contributions, except insomuch as it increases overall GDP.

Figure 5 shows the particular institutions that have contributed the most to the development of the Algorithm Commons.

Interestingly, three private sector organizations make this list of the top 20 contributors: IBM, Bell Labs and the Rand Corporation. Collectively, they account for 11% of the algorithm contributions. But recall, from Figure 1, that only 18% of all contributions come from the

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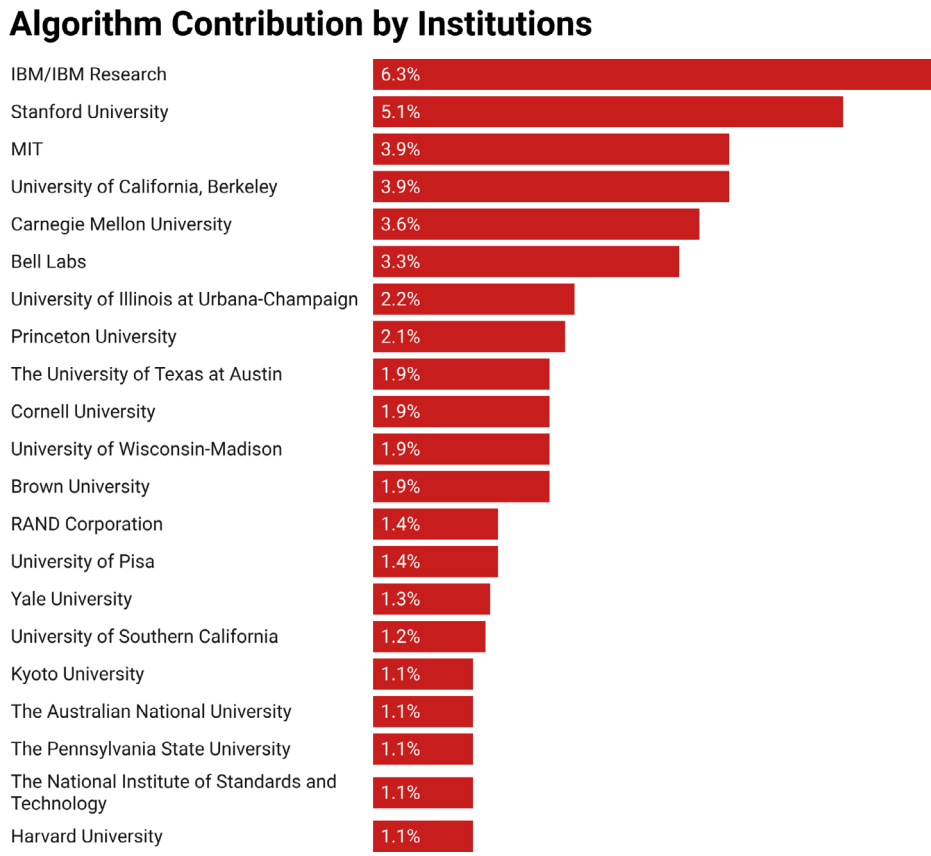

FIGURE 5 Algorithm contributions by institutions

private sector, so these three organizations represent 61% $\left(=\frac{11}{18}\right)$ of all private sector contributions to algorithms. IBM's position as the single largest contributor to algorithm development is particularly noteworthy since the company was also one of the earliest to embrace open-source software, another public good (Capek et al., 2005).

Not surprisingly, the top four public sector contributors in our analysis: Stanford, MIT, UC Berkeley, Carnegie Mellon, are also the four strongest computer science programs in the United States (U.S. News and World Report, 2020).

5 | CONCLUSION

This article analyzes the building of the Algorithm Commons, the set of efficient computer implementations that act as a productivity-enhancer for an enterprise's computing power. We find that the short answer to "who is building the Algorithm Commons" is: the United States, which hosted 64% of the discoverers and where 38% of them were born. Outside the U.S., we find that being a richer country, either in terms of GDP per capita or total GDP, is predictive of making more algorithm contributions. This suggests that algorithm development may exhibit the "lost Einsteins" problem where those with natural talent are stymied because of a lack of opportunity.

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The institutions developing the Algorithm Commons are largely in the public/not-for-profit sector (82%), with the strongest computer science schools contributing disproportionately. Even though the private sector plays a small role overall, several companies are large algorithm contributors, particularly: IBM, Bell Labs, and the Rand Corporation, who collectively represent 61% of all private sector contributions.

Collectively, our work suggests that U.S. enterprises likely benefited significantly from a technology first-mover advantage because the Algorithm Commons was built largely at U.S. universities and in the U.S.' most important private sector research labs. Our findings also provide the first step in understanding how algorithm improvement, a long-overlooked but central part of computing progress, has improved enterprise performance.

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ENDNOTES

[1] Technical note: the clearest examples are when the most efficient algorithm was "exponential" but where a later solution can solve it in "polynomial" time.

[2] We rely on third-party reporting for these examples because we cannot observe the code used by these firms. Absent that code, we cannot know how much of the improvement that they describe should more properly be called software performance engineering rather than algorithm improvement.

[3] This, for example, excludes algorithms that are only approximately correct or benchmarks that are measurable, but where the problem does not have a single "right" answer.

[4] Technically: those that reduced the best asymptotic time complexity of the algorithm family.

[5] Conceptually straight-forward, but computationally inefficient.

[6] Percentages add to more than 100% because of rounding.

[7] Econometric note: implicitly we are assuming that reverse causality is not an issue here. Given the small number of algorithm contributions for nearly all countries, we feel safe in making the assumption that while such improvements may have significant effects for individual firms, they will not be an important driver of GDP per capita or population outside the United States.

[8] Percentages add to less than 100% because of rounding.

[9] Algorithms from the 1940s and earlier are dropped from this graph because a large share of the developments from that period do not have easily-identifiable research papers marking their discovery.

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